Also

$$\begin{aligned}\phi[(a+bi)(c+di)] &= \phi[(ac-bd)+(ad+bc)i] = \begin{bmatrix} ac-bd & ad+bc \\ -ad-bc & ac-bd \end{bmatrix} \\ &= \begin{bmatrix} a & b \\ -b & a \end{bmatrix}\begin{bmatrix} c & d \\ -d & c \end{bmatrix} = \phi(a+bi)\phi(c+di).\end{aligned}$$

Thus ϕ is a homomorphism. It is obvious that ϕ is one to one. Hence ϕ exhibits an isomorpism of $\mathbb{C}$ with the subring $\phi[\mathbb{C}]$, which therefore must be a field.

38. a. For $x, y \in R$, we have

$$\lambda_a(x+y) = a(x+y) = ax + ay = \lambda_a(x) + \lambda_a(y).$$

Thus λ_a is a homomorphism of $\langle R, + \rangle$ with itself, that is, an element of $\text{End}(\langle R, + \rangle)$

b. Note that for $a, b \in R$, we have $(\lambda_a\lambda_b)(x) = \lambda_a(\lambda_b(x)) = \lambda_a(bx) = a(bx) = (ab)x = \lambda_{ab}(x)$. Thus $\lambda_a\lambda_b = \lambda_{ab}$ and R' is closed under multiplication. We also have $(\lambda_a + \lambda_b)(x) = \lambda_a(x) + \lambda_b(x) = ax + bx = (a+b)x = \lambda_{a+b}(x)$, so $\lambda_a + \lambda_b = \lambda_{a+b}$. Thus R' is closed under addition. From what we have shown, it follows that $\lambda_0 + \lambda_a = \lambda_{0+a} = \lambda_a$ and $\lambda_a + \lambda_0 = \lambda_{a+0} = \lambda_a$ so λ_0 acts as additive identity. Finally, $\lambda_{-a} + \lambda_a = \lambda_{-a+a} = \lambda_0$ and $\lambda_a + \lambda_{-a} = \lambda_{a-a} = \lambda_0$ so R' contains an additive inverse of each element. Thus R' is a ring.

c. Let $\phi : R \to R'$ be defined by $\phi(a) = \lambda_a$. By our work in Part(**b**), we see that $\phi(a+b) = \lambda_{a+b} = \lambda_a + \lambda_b = \phi(a) + \phi(b)$, and $\phi(ab) = \lambda_{ab} = \lambda_a\lambda_b = \phi(a)\phi(b)$. Thus ϕ is a homomorphism, and is clearly onto R'. Suppose that $\phi(a) = \phi(b)$. Then $ax = bx$ for all $x \in R$. Because R has unity (and this is the only place where that hypothesis is needed), we have in particular $a1 = b1$ so $a = b$. Thus ϕ is one to one and onto R', so it is an isomorphism.

27. Prime and Maximal Ideals

1. Because a finite integral domain is a field, the prime and the maximal ideals coincide. The ideals $\{0, 2, 4\}$ and $\{0, 3\}$ are both prime and maximal because the factor rings are isomorphic to the fields $\mathbb{Z}_2$ and $\mathbb{Z}_3$ respectively.

2. Because a finite integral domain is a field, the prime and the maximal ideals coincide. The prime and maximal ideals are $\{0, 2, 4, 6, 8, 10\}$ and $\{0, 3, 6, 9\}$ because the factor rings are isomorphic to the fields $\mathbb{Z}_2$ and $\mathbb{Z}_3$ respectively.

3. Because a finite integral domain is a field, the prime and the maximal ideals coincide. The prime and maximal ideals are $\{(0,0), (1,0)\}$ and $\{(0,0), (0,1)\}$ because the factor rings are isomorphic to the field $\mathbb{Z}_2$.

4. A finite integral domain is a field, so prime and maximal ideals coincide. The prime and maximal ideals are $\{(0,0), (0,1), (0,2), (0,3)\}$ and $\{(0,0), (1,0), (0,2), (1,2))\}$ leading to factor rings isomorphic to the field $\mathbb{Z}_2$.

5. By Theorem 27.25, we need only find all values c such that $x^2 + c$ is irreducible over $\mathbb{Z}_3$. Let $f(x) = x^2$. Then $f(0) = 0, f(1) = 1$, and $f(2) = 1$. We must find $c \in \mathbb{Z}_3$ such that $0 + c$ and $1 + c$ are both nonzero. Clearly $c = 1$ is the only choice.

6. By Theorem 27.25, we need only find all values c such that $x^3 + x^2 + c$ is irreducible over $\mathbb{Z}_3$. Let $f(x) = x^3 + x^2$. Then $f(0) = 0, f(1) = 2$, and $f(2) = 0$. We must find $c \in \mathbb{Z}_3$ such that $0 + c$ and $2 + c$ are both nonzero. Clearly $c = 2$ is the only choice.

7. By Theorem 27.25, we need only find all values c such that $g(x) = x^3 + cx^2 + 1$ is irreducible over $\mathbb{Z}_3$. When $c = 0, g(2) = 0$ and when $c = 1, g(1) = 0$, butwhen $c = 2, g(x)$ has no zeros. Thus $c = 2$ is the only choice.

8. By Theorem 27.25, we need only find all values c such that $x^2 + x + c$ is irreducible over $\mathbb{Z}_5$. Let $f(x) = x^2 + x$. Then $f(0) = 0, f(1) = 2, f(2) = 1, f(3) = 2$, and $f(4) = 0$. We must find $c \in \mathbb{Z}_5$ such that $0 + c, 1 + c$,and $2 + c$ are all nonzero. Clearly $c = 1$ and $c = 2$ both work.

9. By Theorem 27.25, we need only find all values c such that $g(x) = x^2 + cx + 1$ is irreducible over $\mathbb{Z}_5$. We compute that when $c = 0, g(2) = 0$, when $c = 1, g(x)$ has no zeros, when $c = 2, g(-1) = 0$, when $c = 3, g(1) = 0$, and when $c = 4, g(x)$ has no zeros. Thus c can be either 1 or 4.

10. The definition is incorrect. We need to specify that the ideals are not R.

A **maximal ideal** of a ring R is an ideal M such that there is no ideal N of R such that $M \subset N \subset R$.

11. The definition is incorrect nonsense.

A **prime ideal** of a commutative ring R is an ideal N such that if $a, b \in R$ and $ab \in N$, then either $a \in N$ or $b \in N$.

12. The definition is correct, although this is not the way it is phrased in the text.

13. The definition is correct, although this is not the way it is phrased in the text.

14. F T T F T T T F T F

15. $\mathbb{Z} \times 2\mathbb{Z}$ is a maximal ideal of $\mathbb{Z} \times \mathbb{Z}$, for the factor ring is isomorphic to $\mathbb{Z}_2$, which is a field.

16. $\mathbb{Z} \times \{0\}$ is a prime ideal of $\mathbb{Z} \times \mathbb{Z}$ that is not maximal, for the factor ring is isomorphic to $\mathbb{Z}$ which is an integral domain, but not a field.

17. $\mathbb{Z} \times 4\mathbb{Z}$ is a proper ideal of $\mathbb{Z} \times \mathbb{Z}$ that is not prime, for the factor ring is isomorphic to $\mathbb{Z}_4$ which has divisors of zero.

18. $\mathbb{Q}/\langle x^2 - 5x + 6\rangle$ is not a field, because $x^2 - 5x + 6 = (x - 2)(x - 3)$ is not an irreducible polynomial, so the ideal $\langle x^2 - 5x + 6\rangle$ is not maximal.

19. $\mathbb{Q}/\langle x^2 - 6x + 6\rangle$ is a field, because the polynomial $x^2 - 6x + 6$ is irreducible by the Eisenstein condition with $p = 2$ or $p = 3$, so $\langle x^2 - 6x + 6\rangle$ is a maximal ideal.

20. If $a + M$ has no multiplicative inverse in R/M, then the principal ideal generated by $a + M$ does not contain $1 + M$, so it is not R/M. Then the inverse image of this ideal under the canonical homomorphism of R into R/M would be an ideal strictly between M and R.

21. If there were an ideal N strictly between the ideal M and the ring R, then its image under the canonical homomorphism of R into R/M would be an ideal of R/M strictly between $\{0 + M\}$ and R/M. This is impossible because there are no nontrivial proper ideals in a field.

22. If F is a field, then the division algorithm can be used to show that every ideal N in $F[x]$ is principal, generated by any element of N of minimum possible degee N.

23. Because a maximal ideal in $F[x]$ is a prime ideal, a factorization of $p(x)$ into two polymials both having degree less than $\deg(p(x))$ would mean that $\langle p(x)\rangle$ would contain a polynomial of degree less than the degree of $p(x)$, which is impossible.

24. Theorem 19.11 shows that every finite integral domain is a field. Let N be a prime ideal in a finite commutative ring R with unity. Then R/N is a finite integral domain, and therefore a field, and therefore N is a maximal ideal.

25. Yes, it is possible; $\mathbb{Z}_2 \times \mathbb{Z}_3$ contains a subring isomorphic to $\mathbb{Z}_2$ and one isomorphic to $\mathbb{Z}_3$.

26. Yes, it is possible; $\mathbb{Z}_2 \times \mathbb{Z}_3$ contains a subring isomorphic to $\mathbb{Z}_2$ and one isomorphic to $\mathbb{Z}_3$.

27. No, it is not possible. Enlarging the integal domain to a field of quotients, we would then have a field containing (up to isomorphism) two different prime fields $\mathbb{Z}_p$ and $\mathbb{Z}_q$. The unity of each of these fields would be a zero of $x^2 - x$, but this polynomial has only one nonzero zero in a field, namely the unity of the field.

28. Let M be a maximal ideal of R and suppose that $ab \in M$ but a is not in M. Let $N = \{ra + m \mid r \in R, m \in M\}$. From $(r_1a + m_1) + (r_2a + m_2) = (r_1 + r_2)a + (m_1 + m_2)$, we see that N is closed under addition. From $r(r_1a + m_1) = (rr_1)a + (rm_1)$ and the fact that M is an ideal, we see that N is closed under multiplication by elements of R, and is of course closed itself under multiplication. Also $0 = 0a + 0$ is in N and furthermore $(-r)a + (-m) = -(ra) - m = -(ra + m)$ is in N. Thus N is an ideal. Clearly N contains M, but $N \neq M$ because $1a + 0 = a$ is in N but a is not in M. Because M is maximal, we must have $N = R$. Therefore $1 \in N$, so $1 = ra + m$ for some $r \in R$ and $m \in M$. Multiplying by b, we find that $b = rab + mb$. But ab and mb are both in M, so $b \in M$. We have shown that if $ab \in M$ and a is not in M, then $b \in M$. This is the definition of a prime ideal.

29. We use the addendum to Theorem 26.3 stated in the final paragraph of Section 26 and proved in Exercise 22 of that section. Suppose that N is any ideal of R. By the addendum mentioned and using the canonical homomorphism $\gamma : R \to R/N$, if M is a proper ideal of R properly containing N, then $\gamma[M]$ is a proper nontrivial ideal of R/N. This shows that if M is not maximal, then R/N is not a simple ring. On the other hand, suppose that R/N is not a simple ring, and let N' be a proper nontrivial ideal of R/N. By the addendum mentioned, $\gamma^{-1}[N']$ is an ideal of R, and of course $\gamma^{-1}[N'] \neq R$ because N' is a proper ideal of R/N, and also $\gamma^{-1}[N']$ properly contains N because N' is nontrival in R/N. Thus $\gamma^{-1}[N']$ is a proper ideal of R that properly contains N, so N is not maximal. We have proved p if and only if q by proving not p if and only if not q.

This exercise is the straightforward analogue of Theorem 15.18 for groups, that is, a maximal ideal of a ring is analogous to a maximal normal subgroup of a group.

30. Every ideal of $F[x]$ is principal by Theorem 26.24. Suppose $\langle f(x)\rangle \neq \{0\}$ is a proper prime ideal of $F[x]$. Then every polynomial in $\langle f(x)\rangle$ has degree greater than or equal to the degree of $f(x)$. Thus if $f(x) = g(x)h(x)$ in $F[x]$ where the degrees of both $g(x)$ and $h(x)$ are less than the degree of $f(x)$, neither $g(x)$ nor $h(x)$ can be in $\langle f(x)\rangle$. This would contradict the fact that $\langle f(x)\rangle$ is a prime ideal, so no such factorization of $f(x)$ in $F[x]$ can exist, that is, $f(x)$ is irreducible in $F[x]$. By Theorem 26.25, $\langle f(x)\rangle$ is therefore a maximal ideal of $F[x]$.

31. If $f(x)$ divides $g(x)$, then $g(x) = f(x)q(x)$ for some $q(x) \in F[x]$, so $g(x) \in \langle f(x)\rangle$ because this ideal consists of all multiples of $f(x)$. Conversely, if $g(x) \in \langle f(x)\rangle$, then $g(x)$ is some multiple $h(x)f(x)$ of $f(x)$ for $h(x) \in F[x]$. The equation $g(x) = h(x)f(x)$ is the definition of $f(x)$ dividing $g(x)$.

32. The equation

$$[r_1(x)f(x) + s_1(x)g(x)] + [r_2(x)f(x) + s_2(x)g(x)] = [r_1(x) + r_2(x)]f(x) + [s_1(x) + s_2(x)]g(x)$$

shows that N is closed under addition. The equation

$$[r(x)f(x) + s(x)(g(x)]h(x) = h(x)[r(x)f(x) + s(x)g(x)] = [h(x)r(x)]f(x) + [h(x)s(x)]g(x)$$

shows that N is closed under multiplication by any $h(x) \in F[x]$; in particular N is closed under multiplication. Now $0 = 0f(x) + 0g(x)$ and $-[r(x)f(x) + s(x)g(x)] = [-r(x)]f(x) + [-s(x)]g(x)$ are in N, so we see that N is an ideal.

Suppose now that $f(x)$ and $g(x)$ have different degrees and that$N \neq F[x]$. Suppose that $f(x)$ is irreducible. By Theorem 26.25, we know that then $\langle f(x)\rangle$ is a maximal ideal of $F[x]$. But clearly $\langle f(x)\rangle \subseteq N$. Because $N \neq F[x]$, we must have $\langle f(x)\rangle = N$. In particular $g(x) \in N$ so $g(x) = f(x)q(x)$. Because $f(x)$ and $g(x)$ have different degrees, we see that $g(x) = f(x)q(x)$ must be a factorization of $g(x)$ into polynomials of smaller degree than the degree of $g(x)$. Hence $g(x)$ is not irreducible.

33. Given that the Fundamental Theorem of Algebra holds, let N be the smallest ideal of $\mathbb{C}[x]$ containing r polynomials $f_1(x), f_2(x), \cdots, f_r(x)$. Because every ideal in $\mathbb{C}[x]$ is a principal ideal, we have $N = \langle h(x)\rangle$ for some polynomial $h(x) \in \mathbb{C}[x]$. Let $\alpha_1, \alpha_2, \cdots, \alpha_s$ be all the zeros in $\mathbb{C}$ of $h(x)$, and let α_i be a zero of multiplicity m_i. By the Fundamental Theorem of Algebra, $h(x)$ must factor into linear factors in $\mathbb{C}[x]$, so that

$$h(x) = c(x - \alpha_1)^{m_1}(x - \alpha_2)^{m_2} \cdots (x - \alpha_s)^{m_s}.$$

Note each α_i is a zero of every $f_j(x)$ because each $f_j(x)$ is a multiple of the generator $h(x)$ of N. Thus by hypothesis, each α_i is a zero of $g(x)$. The Fundamental Theorem of Algebra shows that $g(x) = k(x)(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_s)$ for some polynomial $k(x) \in \mathbb{C}[x]$. Let m be the maximum of $m_1, m_2, \cdots, m_s$. Then $g(x)^m$ has each $(x - \alpha_i)^{m_i}$ as a factor, and thus has $h(x)$ as a factor, so $g(x)^m \in \langle h(x)\rangle = N$.

Conversely, let the Nullstellensatz for $\mathbb{C}[x]$ hold. Suppose that the Fundamental Theorem of Algebra does not hold, so that there exists a nonconstant polynomial $f_1(x)$ in $\mathbb{C}[x]$ having no zero in $\mathbb{C}$. Then every zero of $f_1(x)$ is also a zero of every polynomial in $\mathbb{C}[x]$, because there are no zeros of $f_1(x)$. By the Nullstellensatz for $\mathbb{C}[x]$, every element of $\mathbb{C}[x]$ has the property that some power of it is in $\langle f_1(x)\rangle$, so that some power of every polynomial in $\mathbb{C}[x]$ has $f_1(x)$ as a factor. This is certainly impossible, because $1 \in \mathbb{C}[x]$ and $f_1(x)$ is a nonconstant polynomial and thus is not a factor of $1^n = 1$ for any positive integer n. Thus there can be no such polynomial $f_1(x)$ in $\mathbb{C}[x]$, and the Fundamental Theorem of Algebra holds.

34. **a.** Let $a_1, a_2 \in A$ and $b_1, b_2 \in B$. Then $(a_1 + b_1) + (a_2 + b_2) = (a_1 + a_2) + (b_1 + b_2)$ because addition is commutative. This shows that $A + B$ is closed under addition. For $r \in R$, we know that $ra_1 \in A$ and $rb_1 \in B$, so $r(a_1 + b_1) = ra_1 + rb_1$ is in $A + B$. A similar argument with multiplication on the right shows that $(a_1 + b_1)r = a_1r + b_1r$ is in $A + B$. Thus $A + B$ is closed under multiplication on the left or right by elements of R, in particular, multiplication is closed on $A + B$. Because $0 = 0 + 0 \in A + B$ and $-(a_1 + b_1) = (-a_1) + (-b_1)$ is in $A + B$, we see that $A + B$ is an ideal.

b. Because $a + 0 = a$ is in $A + B$ and $0 + b = b$ is in $A + B$ for all $a \in A$ and $b \in B$, we see that $A \subseteq (A + B)$ and $B \subseteq (A + B)$.

35. **a.** It is clear that AB is closed under addition; [a sum of m products of the form a_ib_i] + [a sum of n products of the form a_jb_k] is a sum of $m + n$ prodcts of this form, and hence is in AB. Because A

and B are ideals, we see that $r(a_ib_i) = (ra_i)b_i$ and $(a_ib_i)r = a_i(b_ir)$ are again of the form a_jb_j. The distributive laws then show that each sum of products a_ib_i when multiplied on the left or right by $r \in R$ produces again a sum of such products. Thus AB is closed under multiplication by elements of R, and hence is closed itself under multiplication. Because $0 = 00$ and $-(a_ib_i) = (-a_i)(b_i)$ are in AB, we see that AB is indeed an ideal.

b.Regarding a_ib_i as a_i in A multiplied on the right by an element b_i of R, we see that a_ib_i is in the ideal A. Regarding a_ib_i as b_i in B multiplied on the left by an element a_i of R, we see that a_ib_i is in B, so $a_ib_i \in A \cap B$. Because A and B are closed under addition, we see that any element of AB is contained in both A and B, so $AB \subseteq A \cap B$.

36. Let $x, y \in A : B$, and let $b \in B$. Then $xb \in A$ and $yb \in A$ for all $b \in B$, so $(x + y)b = xb + yb$ is in A for all $b \in B$, because A is closed under addition. Thus $A : B$ is closed under addition.

Turning to multiplication, let $r \in R$. We want to show that xr and rx are in $A : B$, that is, that $(xr)b$ and $(rx)b$ are in A for all $b \in B$. Because multiplication is commutative by hypothesis, it suffices to show that xbr is in A for all $b \in B$. But because $x \in A : B$, we know that $xb \in A$ and A is an ideal, so $(xb)r \in A$. Thus $A : B$ is closed under multiplication by elements in R; in particular, it is itself closed under multipliction.

Because $0b = 0$ and $0 \in A$, we see that $0 \in A : B$. Because $(-x)b = -(xb)$ and $xb \in A$ implies $(-xb) \in A$, we see that $A : B$ contains the additive identity and additive inverse of each of its elements.

37. Clearly S is closed under addition, contains the zero matrix, and contains the additive inverse of each of its elements. The computation

$$\begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} ac & ad \\ 0 & 0 \end{bmatrix}$$

shows that S is closed under multiplication, so it is a subring of $M_2(F)$. The computations

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ a & b \end{bmatrix}$$

and

$$\begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix} \begin{bmatrix} c & d \\ e & f \end{bmatrix} = \begin{bmatrix} ac + be & ad + bf \\ 0 & 0 \end{bmatrix}$$

show that S is not closed under left multiplication by elements of $M_2(F)$, but is closed under right multiplication by those elements. Thus S is a right deal, but not a left ideal, of $M_2(F)$.

38. Let $R = M_2(\mathbb{Z}_2)$. The computations

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix} \text{ and } \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & a \\ d & c \end{bmatrix}$$

show that for every matrix in an ideal N of R, the matrix obtained by interchanging its rows and the matrix obtained by interchanging its columns are again in N. Thus if N contains any one of the four matrices having 1 for one entry and 0 for all the others, then N contains all four such matrices, and hence all nonzero matrices because any matrix in R is a sum of such matrices and N is closed under addition. By interchanging rows and columns, every nonzero matix with at least two nonzero entries can be brought to one of the following forms: